

Ashi Poorey

apoorey2@illinois.edu | in [ashipoorey](#) | [github](#) | [personal website](#)

Champaign-Urbana, IL | +1 217-530-8422

EDUCATION

Bachelor of Science - Astrophysics

University of Illinois Urbana Champaign

Minors - Data Science and Computational Science & Engineering

2027

RESEARCH & INDUSTRY EXPERIENCE

Space Astronomy Summer Program intern - STScI

Jun 2026 - Aug 2026

Undergraduate Researcher [REU]

- Developed and trained machine learning models to identify rare galaxies exhibiting broad Mg II emission in Sloan Digital Sky Survey spectra, targeting signatures of fading supermassive black holes
- Analyzed large spectroscopic datasets (~1.4 million sources) to isolate and validate a small sample of candidate objects, enabling study of black hole accretion duty cycles
- Designed scalable methods for deployment on future surveys (SDSS extensions, DESI, etc.) to discover and characterize transient active galactic nuclei in time-domain astronomy

UIUC Team Gaia - Prof Jiayin Dong

Aug 2025 - present

Undergraduate Researcher

- Developed an exoplanet detection pipeline using ESA Gaia mission data, reconstructing existing Bayesian inference code from PyStan to PyTensor + PyMC.
- Applied statistical methods to time-series astrometric data for exoplanet detection, with Bayesian statistics and probabilistic modeling
- Led research paper discussions and participated in hackathon-style coding sessions, strengthening skills in Python programming, scientific problem-solving, and teamwork.

UIS Stellar Evolution Research - Prof John Martin

Feb 2026 - present

Undergraduate Researcher

- Conducted research on Warm Hypergiants, investigating their variability and evolutionary behavior
- Analyzed time-series photometric features using Lomb–Scargle periodograms to identify periodic and quasi-periodic variability in M31 and M33 massive stars
- Interpreted variability trends in the context of stellar pulsations, mass loss, and late-stage massive star evolution; contributing to ongoing analysis for presentation and publication

QuantCAD - Duality Cohort & Chicago Quantum Exchange group

Jun 2025 - Aug 2025

Quantum Physics Intern (NASA-backed projects)

- Worked with PI on quantum device irradiation research, contributing to simulation-driven analysis, experimental planning, and technical proposal writing for national lab facilities
- Conducted data analysis in optics experiments and evaluating quantum signal response under varying magnetic fields; along with research related to NV center quantum sensing
- Designed and modeled Halbach array magnet configurations, integrating physics simulations with feasibility research to inform experimental setups

UIUC Divergence 2% Lab - Optical Computing project
Undergraduate Researcher

Jan 2025 - May 2025

- Contributed to the simulation and optimization of photonic metasurfaces, integrating physical theory with computational design
- Conducted theoretical research and extensive literature review on photonic logic, optical computing, fourier mathematics, and physics

Illinois Space Society
NASA RASC-al Team

Aug 2024 - May 2025

- Researched cryogenic fluid management systems and in-situ resource utilization for long-duration lunar missions
- Engaged in mission architecture and design analysis for human exploration concepts, as part of the Chemical Systems Sub-Team
- Collaborated with 15+ members across sub-teams to refine mission concept, perform feasibility analysis, and prepare technical deliverables

PROGRAMS & TRAINING

STScI JWST Summer School – High-Redshift Transients and Time Domain Astronomy *Aug 2025*

- Selected to participate in a program combining lectures on high-z transient science with hands-on training in JWST observation planning, calibration pipelines, and data analysis.
- Gained practical experience in creating JWST observing programs and processing early-universe astrophysical data using advanced infrared techniques.
- Completed daily hands-on assignments involving JWST proposal design, calibration pipeline execution, and analysis of high-z transient datasets to produce scientifically interpretable results

NASA L'SPACE – Mission Concept Academy

Sep 2024 - Dec 2024

- **Project Manager** for a 12-member interdisciplinary team to develop a mission architecture aligned with NASA's Lucy Mission and planetary science goals
- Coordinated weekly meetings, delegated technical tasks, and integrated deliverables into a 50+ page final report scoring among the top 10%
- Presenting progress updates in formal briefings and design reviews to NASA mentors

CONFERENCES, TALKS & PRESENTATIONS

1. **APS CU*IP 2026** - RUWE-Based Pipeline for Inferring Planetary Companions from Gaia DR3 [*poster*]
2. **CAURS 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 [*poster*]
3. **UIUC URS 2026** - RUWE-Based Pipeline for Inferring Planetary Companions from Gaia DR3 [*poster*]
4. **UIUC Astrofest 2026** - Photometric Variability of Warm Hypergiants in M31 and M33 [*poster*]
5. **UIUC Physics Seminar 2025** - NV Centers & All-Electric Magnetometry [*talk/presentation*]
6. **CQE Quantum Sensing Summer Program** with NSF QuBBE 2025 [*talk/panel*]
7. **Tech Chicago Week** Intern Event 2025 [*invited attendee*]

AWARDS

1. **Celia Mathews Elliot Prize** for Writing in Science and Engineering - *UIUC Department of Physics*
2. **International Astronomy and Astrophysics Competition** Senior Qualifier - Pre-Final Round

OUTREACH, VOLUNTEERING & TEACHING

The Jeffries Academic Center - *Math & Physics Tutor* *Jan 2025 - May 2025*
Provided weekly academic support in Physics and Math to over five students

UIUC College of Liberal Arts & Sciences - *Student Ambassador* *Sep 2025 - present*
Representing the college at outreach events and assisting prospective students with academics, campus life, and opportunities in LAS

UIUC LAS Lift Off 2025 - *Department of Astronomy booth* *Aug 2024*
Represented the Astronomy department, welcoming new students and answering questions about courses, research opportunities, and campus resources

The Illinois Student Undergraduate Research Journal - *Content Editor* *Aug 2024 - present*
Interviewed faculty and graduate researchers to write feature articles, oversaw a team of editors and managing 10+ journal submissions per semester

Undergraduate Pre-PhD Society - *Secretary* *June 2025 - Dec 2025*
Coordinated communications, organizing meetings, and supporting initiatives to prepare students for graduate research careers

LAS 100 - College of Liberal Arts & Sciences - *Class Intern* *Aug 2024 - Oct 2024*
Mentored 7 international students by facilitating weekly discussions, reviewing and grading assignments, and providing transition support

Illinois Space Society - *Peer Mentor* *Aug 2024 - Dec 2024*
Mentored a freshman student by providing guidance on academics, opportunities, and career advice

TEDxUIUC - *Logistics & Event Planning* *Jan 2024 - Dec 2024*
Coordinated logistics and cross-team planning, managing venue setup, speaker arrangements, and event execution to ensure a seamless large-scale event experience

Engineering Open House - Exhibit Volunteer

April 2025

Led an interactive exhibit on space habitat design, guiding participants through challenges of building on Mars and facilitating hands-on prototyping using the engineering design process

PROJECTS

Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics | Publication

Co-authored research on diversity and retention in astrophysics, synthesizing literature and community insights into actionable recommendations

NASA RASC-al: MATER: Mars Architecture for Technology Evaluation and Research | Proposal

Co-developed a science-driven Mars mission architecture incorporating advanced technology demonstrators for human exploration

Radio Astronomy Capstone Project: S255IR NIRS3 (ALMA Data)

Processed ALMA interferometric data using CASA and CARTA to produce continuum and C³⁴S molecular line results, generating publication-quality figures and deriving astrophysical interpretations consistent with observational studies

ASTR 401 Final Paper: Hawking Radiation & the Information Paradox

Conducted theoretical analysis of modern approaches to resolving the black hole information paradox through literature synthesis

STAT 107 project: Stellar Population Analysis

Applied data science techniques to construct and analyze a Hertzsprung–Russell diagram, investigating stellar evolution trends

Illinois Data Science Club: Data Dive Project

Used exploratory data analysis and logistic regression to investigate factors affecting animal survival rates in shelters, focusing on the impact of intake reasons, and presented it to judges from John Deere and P&G

TECHNICAL SKILLS

Languages: Python, C++, SQL, MATLAB, Linux, Arduino

Libraries: AstroPy, NumPy, SciPy, Pandas, Seaborn, Statsmodel, MagPy, PyStan, PyTensor, Scikit